Principles of Microeconomics — Week 3, Day 3

In-Class Problem Set: Externalities, Public Goods, the Commons

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This problem set accompanies the Day 3 session. Numbers differ from the lecture examples. Solutions (instructor copy) are collected at the end; set `\solnfalse` in the preamble to produce a student version.

Exercise 1 — A negative externality

A competitive market for a fertiliser has demand and private marginal cost

$$P = 90 - Q \quad \text{(marginal benefit)}, \qquad MPC = 10 + Q,$$

and each unit produced causes pollution worth a constant marginal external cost $MEC = 20$.

(a) Find the market (private) equilibrium quantity.

(b) Write the marginal social cost MSC and find the socially efficient quantity Q^*.

(c) Compute the deadweight loss of the unregulated market.

(d) What per-unit corrective (Pigouvian) tax restores efficiency, and how much revenue does it raise?

Exercise 2 — A public good and a common resource

Part A — Public good. A shared flood wall of size Q protects two farms, with

$$MB_A = 50 - Q, \qquad MB_B = 30 - Q, \qquad MC = 40.$$

(a) Find the efficient size Q^* by adding the two marginal benefits (vertically) and setting the sum equal to MC.

(b) If each farm contributes voluntarily (up to where its *own* marginal benefit equals cost), how much is built in total? Who free-rides? Compare with Q^*.

Part B — Common fishery. B boats fish a common ground. The value per boat is $120 - 2B$ and the cost per boat is 40.

(c) Find the number of boats a *single owner* would choose (the efficient B^*), where the *extra* boat's value equals its cost.

(d) Find the open-access number of boats B_{oa}, where the *average* value per boat equals the cost. By how much is the fishery over-used?

Reconvene: in each case state the efficient quantity, the market outcome, and the one signal the price is missing.